Assignment 5 - Dataset Report

Kendall Beaver

2024-02-17

Question

What are some of the key characteristics and trends of people who take out home mortgages?

Key Characteristics & Trends of Mortgage Borrowers

After analyzing U.S. residential mortgage data from the **National Mortgage Database** (NMDB®) from 1998-2021, this report has concluded that interest rates have a large influence on a borrower's decision to take out a mortgage and therefore, creates an indirect relationship between borrowers and mortgages (Figure 1a).

Additionally, there have always been more borrowers of refinance loans than new home loans, indicating that existing borrowers constantly monitor and take advantage of lower interest rates and thus loan costs, or will choose to cash some of their home equity to finance various endeavours regardless of interest rates (Figure 1b).

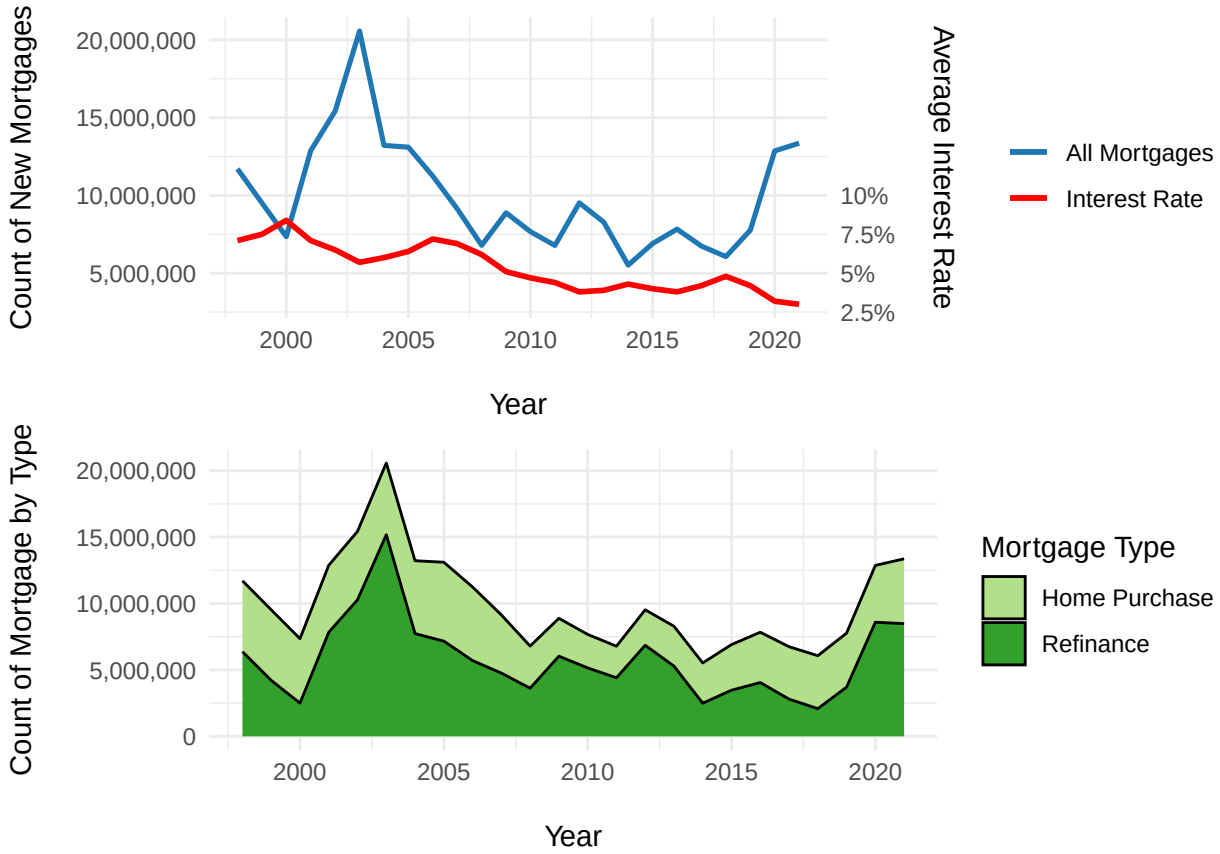


Figure 1a (top plot)**.** Line plot of mortgage count vs. average interest rate per year.
Figure 1b (bottom plot)**.** Stacked area chart of mortgage type–home purchase vs. refinance–per year.

Age & Credit Score

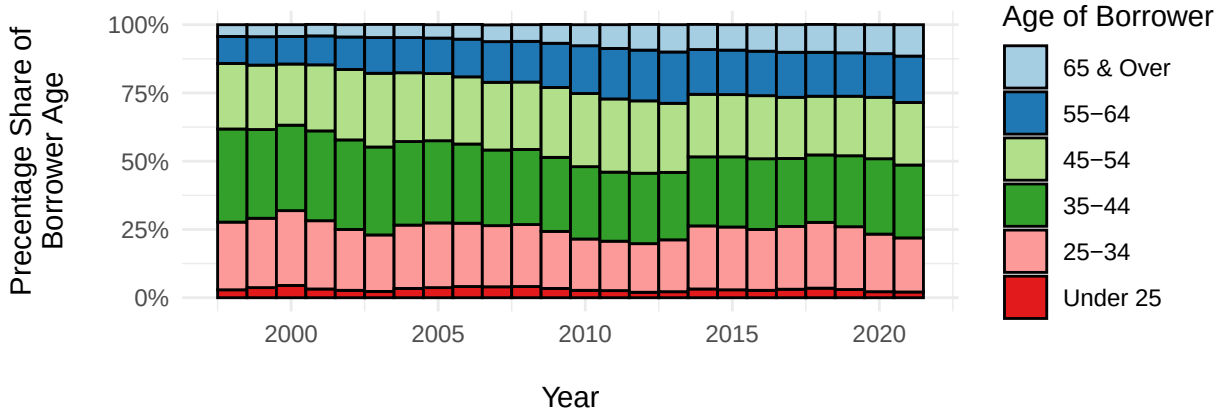
This report then assessed the percentage share of borrowers' age and average credit score, and out of the six age groups presented in Table 1 and Figure 2a, only two age groups increased in size while all other age groups decreased, "65 & Over" and "55-64"; however, they do not make up a majority percentage share of borrowers' age, that has been and remains the ages of 25-54 ("25-34", "35-44" & "45-54").

Age of Borrower	1998	2021	Rate of Change
65 & Over	4.3%	11.5%	167.4%
55-64	9.9%	17.0%	71.7%
45-54	24.0%	22.9%	-4.6%
35-44	34.1%	26.7%	-21.7%
25-34	24.8%	19.8%	-20.2%
Under 25	2.9%	2.1%	-27.6%

Table 1. Percentage share of age categories based on year ends of dataset and rate of change.

The trend was the same for credit scores (Figure 2b), where the two high-end credit scores, "Excellent (781-850)" and "Good (661-780)", increased by 28.6% and 19.4%, respectively; and these scores also account for the largest share of this category: 33.3% and 55.5%, respectively.

While borrowers, on average, have become more older, the trend remains clear that despite age, lenders have become more risk adverse and are willing to originate mortgages for borrowers who have good or excellent credit rather than borrowers who have anything less than these ratings.



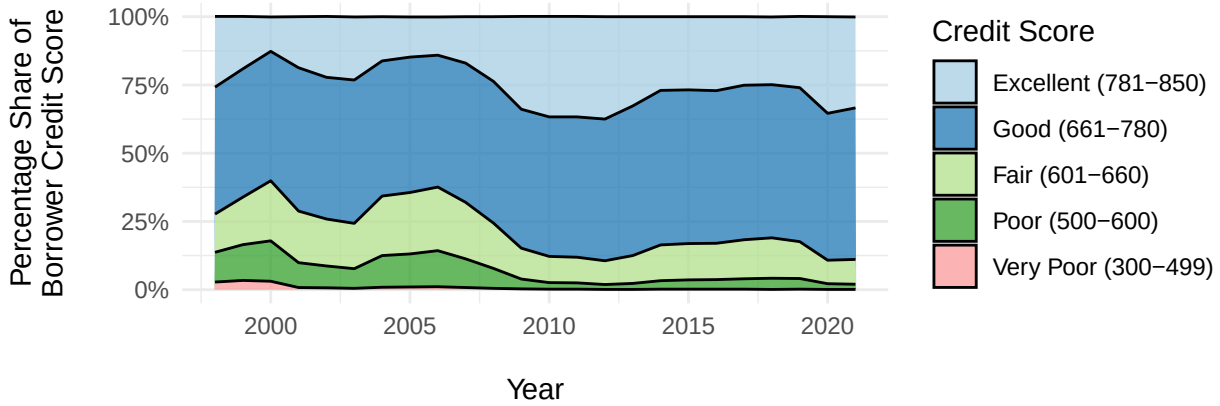
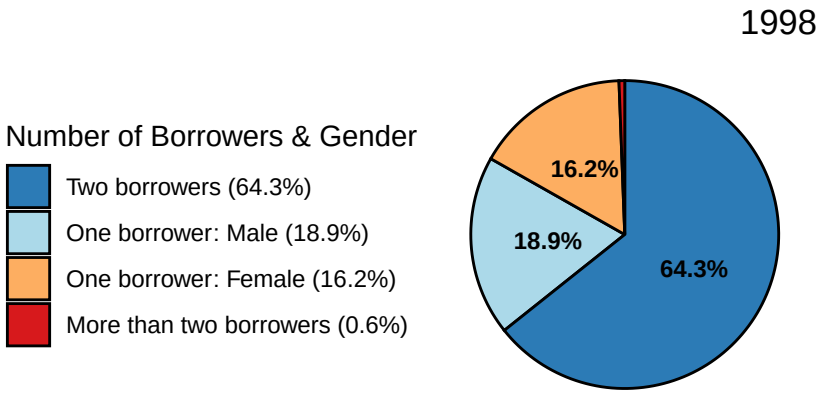


Figure 2a (top plot)**.** Stacked bar plot of borrowers' age by percentage share.
Figure 2b (bottom plot)**.** Stacked area chart of borrowers' credit score by percentage share.

Gender & Number of Borrowers

Lastly, this report analyzed the relationship between the gender and number of borrowers per mortgage, where "two borrowers" have remained the primary type of borrower (though it's not explicitly stated in the dataset, "two borrowers" are assumed to be one male and one female); yet their presence has decreased by 31% from 64.3% in 1998 (Figure 3a), to 44.4% in 2021 (Figure 3b), and rising in dominance is the single male borrower at 33.5%, in 2021, followed by the single female borrower at 21.4% during the same year.

The single male and female borrower combined have already overtaken half the market at 54.9% in 2021, which is almost as high as two borrowers' share in 1998, at 64.3%, and is a number that they may soon overtake.



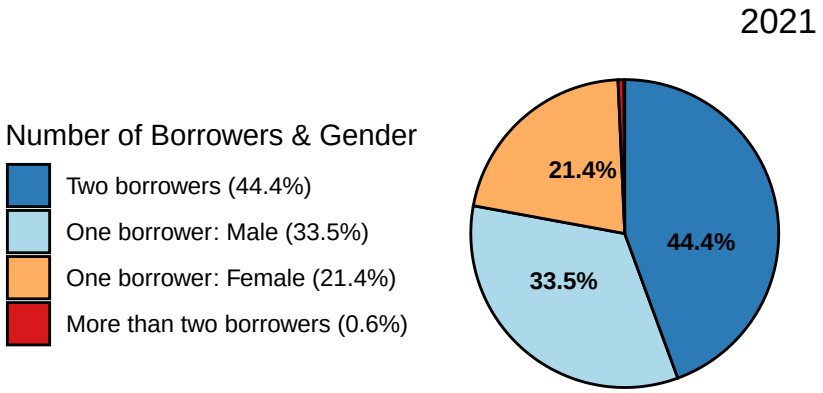


Figure 3a (top plot). Pie chart of gender and number of borrowers in 1998.
Figure 3b (bottom plot). Pie chart of gender and number of borrowers in 2021.

Conclusion

The trend and characteristics of the typical borrower today is much different than it was twenty-three-years ago, which was typically composed of a traditional, two-borrower couple or family. In contrast, the average borrower today is most likely older, has a good chance of being a single male borrower—or even a single female borrower—just as much as being two borrowers, and has a good or excellent credit score.

The average borrower today seems to be waiting to take advantage of a good savings or investment opportunity and will refinance their mortgage to do so, making mortgage borrowers in the U.S. the most sophisticated, socially independent, and investment-minded borrowers than ever before—a trend that seems likely to only increase over time.

Code

```r
### Install packages & load libraries

if (!require(pacman))
  install.packages("pacman")

pacman::p_load(dplyr, ggplot2, gridExtra, lubridate, reshape2,
               scales, stringr, forcats, kableExtra, ggrepel)

library(dplyr)
library(ggplot2)
library(gridExtra)
library(lubridate)
library(reshape2)
library(scales)
library(stringr)
library(tidyr)
library(forcats)
library(kableExtra)
library(ggrepel)

mortgages <- read.csv("mortgages.csv")

# Let's make the columns more human readable/friendly
mortgages <- mortgages %>%
  rename(SERIES_ID = SERIESID,
         GEO_LEVEL = GEOLEVEL,
         GEO_ID = GEOID,
         GEO_NAME = GEONAME)

unique(mortgages$GEO_LEVEL)
# National
# Census Region
# Census Division
# State

# Let's clean this up and turn the numeric values into numbers I can understand
mortgages %>%
  mutate(RELATIVE_VALUE = paste0(gsub("\\D", "", VALUE), "%"))

mortgages <- mortgages %>%
  filter(GEO_LEVEL == "National") %>%
  mutate(RELATIVE_VALUE = case_when(
    str_detect(SERIES_ID, "TOT_ORIG") ~ paste0(comma(VALUE * 1000)),
    str_detect(SERIES_ID, "AVE_LOANAMT") |
      str_detect(SERIES_ID, "AVE_PROPVAL") ~ paste0("$", comma(VALUE * 1000)),
    str_detect(SERIES_ID, "PCT") |
      str_detect(SERIES_ID, "DTI") |
      str_detect(SERIES_ID, "RATE") |
      str_detect(SERIES_ID, "LTV") |
      str_detect(SERIES_ID, "AVE_TERM") ~ paste0(format(VALUE, nsmall =
                                                        ifelse(floor(VALUE) == VALUE, 0, 1)), "%"),
    str_detect(SERIES_ID, "AVE_AGE_BORROWER") ~ paste0(round(VALUE), " years old"),
```

```r
    str_detect(SERIES_ID, "AVE_VANTAGESCR") ~ paste0(VALUE, " credit score"),
    TRUE ~ as.character(VALUE)))
```

```r
### AREA PLOT + LINE PLOT: 'All Mortgages'

home_purchases_vs_refinances <- mortgages %>%
    filter((MARKET == "All Mortgages") & (SERIES_ID == "TOT_ORIG"))

interest_rates <- mortgages %>%
    filter((MARKET == "All Mortgages") & (SERIES_ID == "AVE_INTRATE"))

# Plot with all mortgages

home_purchases_vs_refinances <- mortgages %>%
    filter((MARKET == "All Mortgages") & (SERIES_ID == "TOT_ORIG"))

interest_rates <- mortgages %>%
    filter((MARKET == "All Mortgages") & (SERIES_ID == "AVE_INTRATE"))

# Plot with all mortgages

p1 <- ggplot(home_purchases_vs_refinances, aes(x = YEAR, y = VALUE)) + geom_line(aes(color = "All Mortga
    alpha = 1, linewidth = 0.9) + labs(x = "\nYear", y = "Count of New Mortgages\n") +
    scale_y_continuous(labels = function(x) comma(x * 1000)) + theme_minimal()

# Line plot of interest rate
p2 <- ggplot(interest_rates, aes(x = YEAR, y = VALUE)) + geom_line(aes(color = "Interest Rate",
    fill = "Average Interest Rate"), linewidth = 0.5) + labs(y = "Interest Rate") +
    theme_minimal()

# Now let's overlap/combine both plots
p_combined <- p1 + geom_line(data = interest_rates, aes(y = VALUE * 1000, color = "Interest Rate"),
    linewidth = 1) + scale_y_continuous(labels = function(x) comma(x * 1000), sec.axis = sec_axis(~./100
    name = "Average Interest Rate\n", labels = function(x) paste0(x, "%"), breaks = seq(0,
        10, by = 2.5))) + scale_color_manual(values = c(`All Mortgages` = "#1f78b4",
    `Interest Rate` = "red")) + guides(color = guide_legend(title = NULL)) + guides(fill = guide_legend

### SUPPLEMENTAL STACKED AREA PLOT: 'Home Purchase' and 'Refinance'

# Split up 'All Mortgages' into 'Home Purchase' and 'Refinance'
home_purchases_vs_refinances <- mortgages %>%
    filter((MARKET == "All Mortgages (Home Purchase)" | MARKET == "All Mortgages (Refinance)") &
        (SERIES_ID == "TOT_ORIG"))

# Need to rename MARKET IDs so characters are readable in my legend
home_purchases_vs_refinances$MARKET[grepl("Home Purchase", home_purchases_vs_refinances$MARKET)] <- "Hom
home_purchases_vs_refinances$MARKET[grepl("Refinance", home_purchases_vs_refinances$MARKET)] <- "Refinan

# Let's plot the area plot
p3_home_purchases <- ggplot(home_purchases_vs_refinances, aes(x = YEAR, y = VALUE,
    fill = MARKET)) + geom_area(stat = "identity", alpha = 1, col = "black", linewidth = 0.5) +
    labs(x = "\nYear", y = "Count of Mortgage by Type\n", fill = "Mortgage Type") +
    scale_fill_manual(values = c("#b2df8a", "#33a02c")) + scale_y_continuous(labels = function(x) comma
    1000)) + theme_minimal()
```

```r
### COMBINE PLOTS

# combined_plots <- grid.arrange(p_combined, p3_home_purchases)
```

```r
# STACKED BAR PLOT: Credit score

# Try "Set1", otherwise stick to:
# scale_fill_brewer(palette = "Paired")

mortgages_credit_score <- mortgages %>%
  filter((MARKET == "All Mortgages") &
           (SERIES_ID == "PCT_VS_VERYPOOR" |
              SERIES_ID == "PCT_VS_POOR" |
              SERIES_ID == "PCT_VS_FAIR" |
              SERIES_ID == "PCT_VS_GOOD" |
              SERIES_ID == "PCT_VS_EXCELLENT")) %>%
  mutate(DECIMAL_VALUE = VALUE * 0.01) %>%
  mutate(SERIES_ID = case_when(
    SERIES_ID == "PCT_VS_VERYPOOR" ~ "Very Poor (300-499)",
      SERIES_ID == "PCT_VS_POOR" ~ "Poor (500-600)",
      SERIES_ID == "PCT_VS_FAIR" ~ "Fair (601-660)",
      SERIES_ID == "PCT_VS_GOOD" ~ "Good (661-780)",
      SERIES_ID == "PCT_VS_EXCELLENT" ~ "Excellent (781-850)",
      TRUE ~ SERIES_ID))

mortgages_credit_score <- mortgages_credit_score %>%
  mutate(SERIES_ID = fct_relevel(SERIES_ID, "Excellent (781-850)", "Good (661-780)", "Fair (601-660)",

# Stacked Area Plot
stacked_plot <- ggplot(mortgages_credit_score, aes(x = YEAR, y = DECIMAL_VALUE, fill = SERIES_ID)) +
  geom_area(alpha = .75, col = "black", linewidth = .5) +
  labs(x = "\nYear",
       y = "Percentage Share of Average Borrower Credit Score\n",
       fill = "Credit Score") +
  scale_fill_brewer(palette = "Paired") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()

# Table: Change in Percentage of Credit Score from 1998 - 2021

# Create a table for comparison of actual values
mortgages_credit_score %>%
  filter(YEAR == "1998" | YEAR == "2021") %>%
  group_by(SERIES_ID, YEAR) %>%
  summarize(VALUE = list(VALUE)) %>%
  pivot_wider(names_from = YEAR, values_from = VALUE) %>%
  arrange(SERIES_ID) %>%
  rename("Credit Score" = SERIES_ID) %>%
  unnest(c("1998", "2021")) %>%
  mutate(rate_of_change = (`2021` - `1998`) / `1998` * 100) %>%
  mutate("1998" = paste0(format(round(`1998`, 1), nsmall = 1), "%"),
         "2021" = paste0(format(round(`2021`, 1), nsmall = 1), "%"),
         "Rate of Change" = sprintf("%.1f%%", rate_of_change)) %>%
```

```r
  select(-rate_of_change)
```

```r
### PLOT: Percentage Stacked Area Chart (100%) of Borrower Age
mortgages_age_percentages <- mortgages %>%
  filter((MARKET == "All Mortgages") &
           (SERIES_ID == "PCT_AGE_LT25" |
              SERIES_ID == "PCT_AGE_2534" |
              SERIES_ID == "PCT_AGE_3544" |
              SERIES_ID == "PCT_AGE_4554" |
              SERIES_ID == "PCT_AGE_5564" |
              SERIES_ID == "PCT_AGE_GE65")) %>%
  mutate(DECIMAL_VALUE = VALUE * 0.01) %>%
  mutate(SERIES_ID = case_when(
    SERIES_ID == "PCT_AGE_LT25" ~ "Under 25",
    SERIES_ID == "PCT_AGE_2534" ~ "25-34",
    SERIES_ID == "PCT_AGE_3544" ~ "35-44",
    SERIES_ID == "PCT_AGE_4554" ~ "45-54",
    SERIES_ID == "PCT_AGE_5564" ~ "55-64",
    SERIES_ID == "PCT_AGE_GE65" ~ "65 & Over",
    TRUE ~ SERIES_ID))

# Convert SERIES_ID to factor with ordered levels so that I can organize in descending order
mortgages_age_percentages$SERIES_ID <- factor(mortgages_age_percentages$SERIES_ID,
                                              levels = c("65 & Over", "55-64", "45-54", "35-44"
                                                         , "25-34", "Under 25"), ordered = TRUE)

# Bar plot (test)
bar_plot <- ggplot(mortgages_age_percentages, aes(x = YEAR, y = DECIMAL_VALUE, fill = SERIES_ID)) +
  geom_bar(stat = "identity", position = "stack", col = "black", linewidth = 0.5, width = 1) +
  labs(x = "\nYear",
       y = "Precentage Share of Borrower Age\n",
       fill = "Age of Borrower") +
  scale_fill_brewer(palette = "Paired") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()

# Table adapted from previous plot, need to adjust this:
mortgages_age_percentages %>%
  filter(YEAR == "1998" | YEAR == "2021") %>%
  group_by(SERIES_ID, YEAR) %>%
  summarize(VALUE = list(VALUE)) %>%
  pivot_wider(names_from = YEAR, values_from = VALUE) %>%
  arrange(SERIES_ID) %>%
  rename("Age of Borrower" = SERIES_ID) %>%
  unnest(c("1998", "2021")) %>%
  mutate(rate_of_change = (`2021` - `1998`) / `1998` * 100) %>%
  mutate("1998" = paste0(format(round(`1998`, 1), nsmall = 1), "%"),
         "2021" = paste0(format(round(`2021`, 1), nsmall = 1), "%"),
         "Rate of Change" = sprintf("%.1f%%", rate_of_change)) %>%
  select(-rate_of_change)

# Table 3. Average Percentage Share of Borrowers' Age and Rate of Change.
```

```r
### PLOT 4: Pie charts of Borrowers' Race (1998 & 2021)

# Segment 1998 data
mortgages_race_1998 <- mortgages %>%
    filter((MARKET == "All Mortgages") & (YEAR == 1998) & (SERIES_ID == "PCT_WHT" |
        SERIES_ID == "PCT_BLK" | SERIES_ID == "PCT_ASN" | SERIES_ID == "PCT_HPI" |
        SERIES_ID == "PCT_AMI" | SERIES_ID == "PCT_MIX")) %>%
    mutate(DECIMAL_VALUE = VALUE * 0.01) %>%
    mutate(SERIES_ID = case_when(SERIES_ID == "PCT_WHT" ~ "White", SERIES_ID == "PCT_BLK" ~
        "Black or African American", SERIES_ID == "PCT_ASN" ~ "Asian", SERIES_ID ==
        "PCT_HPI" ~ "Hawaiian or Pacific Islander", SERIES_ID == "PCT_AMI" ~ "American Indian or Alaska
        SERIES_ID == "PCT_MIX" ~ "More than one race was reported\nindividually or per the borrowers",
        TRUE ~ SERIES_ID))

# Segment 2021 data
mortgages_race_2021 <- mortgages %>%
    filter((MARKET == "All Mortgages") & (YEAR == 2021) & (SERIES_ID == "PCT_WHT" |
        SERIES_ID == "PCT_BLK" | SERIES_ID == "PCT_ASN" | SERIES_ID == "PCT_HPI" |
        SERIES_ID == "PCT_AMI" | SERIES_ID == "PCT_MIX")) %>%
    mutate(DECIMAL_VALUE = VALUE * 0.01) %>%
    mutate(SERIES_ID = case_when(SERIES_ID == "PCT_WHT" ~ "White", SERIES_ID == "PCT_BLK" ~
        "Black or African American", SERIES_ID == "PCT_ASN" ~ "Asian", SERIES_ID ==
        "PCT_HPI" ~ "Hawaiian or Pacific Islander", SERIES_ID == "PCT_AMI" ~ "American Indian or Alaska
        SERIES_ID == "PCT_MIX" ~ "More than one race was reported\nindividually or per the borrowers",
        TRUE ~ SERIES_ID))

# Create decimal values to show in pie charts
mortgages_race_1998 <- data.frame(SERIES_ID = c("White (88.2%)", "Black or African American (5.2%)",
    "Asian (2.6%)", "Hawaiian or Pacific Islander (0.1%)", "American Indian or Alaska Native (0.2%)",
    "More than one race was reported\nindividually or per the borrowers (3.6%)"),
    DECIMAL_VALUE = c(0.882, 0.052, 0.026, 0.001, 0.002, 0.036), label = paste0(round(c(0.882,
        0.052, 0.026, 0.001, 0.002, 0.036) * 100, 1), "%"))

mortgages_race_2021 <- data.frame(SERIES_ID = c("White (81.1%)", "Black or African American (7.0%)",
    "Asian (3.8%)", "Hawaiian or Pacific Islander (0.2%)", "American Indian or Alaska Native (0.4%)",
    "More than one race was reported\nindividually or per the borrowers (7.5%)"),
    DECIMAL_VALUE = c(0.811, 0.07, 0.038, 0.002, 0.004, 0.075), label = paste0(round(c(0.811,
        0.07, 0.038, 0.002, 0.004, 0.075) * 100, 1), "%"))

# 1998 Pie Chart
p1998 <- ggplot(mortgages_race_1998, aes(x = "", y = DECIMAL_VALUE, fill = SERIES_ID)) +
    geom_bar(stat = "identity", width = 0.5, color = "black") + geom_text(aes(label = ifelse(rank(-DECI
    2, label, "")), position = position_fill(vjust = 0.4), size = 3, fontface = "bold") +
    coord_polar(theta = "y") + labs(x = NULL, y = NULL, fill = "Borrowers' Race") +
    scale_fill_brewer(palette = "RdYlBu") + theme_void() + ggtitle("1998")

# 2021 Pie Chart
p2021 <- ggplot(mortgages_race_2021, aes(x = "", y = DECIMAL_VALUE, fill = SERIES_ID)) +
    geom_bar(stat = "identity", width = 0.5, color = "black") + geom_text(aes(label = ifelse(rank(-DECI
    2, label, "")), position = position_fill(vjust = 0.5), size = 3, fontface = "bold") +
    coord_polar(theta = "y") + labs(x = NULL, y = NULL, fill = "Borrowers' Race") +
    scale_fill_brewer(palette = "RdYlBu") + theme_void() + ggtitle("2021")
```

```r
# Display both pie charts together grid.arrange(p1998, p2021, ncol = 1)
```

```r
### PLOT 5: Pie Charts of Number of Borrowers & Gender (1998 & 2021)

# Segment 1998 data
mortgages_borrowers_gender_1998 <- mortgages %>%
  filter((MARKET == "All Mortgages") &
           (YEAR == 1998) &
           (SERIES_ID == "PCT_MALEBOR" |
              SERIES_ID == "PCT_FEMALEBOR" |
              SERIES_ID == "PCT_TWOBOR" |
              SERIES_ID == "PCT_MULTIBOR")) %>%
  mutate(DECIMAL_VALUE = VALUE * 0.01) %>%
  mutate(SERIES_ID = case_when(
    SERIES_ID == "PCT_MALEBOR" ~ "One borrower: Male (18.9%)",
      SERIES_ID == "PCT_FEMALEBOR" ~ "One borrower: Female (16.2%)",
      SERIES_ID == "PCT_TWOBOR" ~ "Two borrowers (64.3%)",
      SERIES_ID == "PCT_MULTIBOR" ~ "More than two borrowers (0.6%)",
      TRUE ~ SERIES_ID))

# Segment 2021 data
mortgages_borrowers_gender_2021 <- mortgages %>%
  filter((MARKET == "All Mortgages") &
           (YEAR == 2021) &
           (SERIES_ID == "PCT_MALEBOR" |
              SERIES_ID == "PCT_FEMALEBOR" |
              SERIES_ID == "PCT_TWOBOR" |
              SERIES_ID == "PCT_MULTIBOR")) %>%
  mutate(DECIMAL_VALUE = VALUE * 0.01) %>%
  mutate(SERIES_ID = case_when(
    SERIES_ID == "PCT_MALEBOR" ~ "One borrower: Male",
      SERIES_ID == "PCT_FEMALEBOR" ~ "One borrower: Female",
      SERIES_ID == "PCT_TWOBOR" ~ "Two borrowers",
      SERIES_ID == "PCT_MULTIBOR" ~ "More than two borrowers",
      TRUE ~ SERIES_ID))

# Create decimal values to put in 1998 pie chart
mortgages_borrowers_gender_1998 <- data.frame(
  SERIES_ID = c("One borrower: Male (18.9%)", "One borrower: Female (16.2%)", "Two borrowers (64.3%)",
  DECIMAL_VALUE = c(0.189, 0.162, 0.643, 0.006),
  label = paste0(round(c(0.189, 0.162, 0.643, 0.006) * 100, 1), "%")
)

# Create decimal values to put in 2021 pie chart
mortgages_borrowers_gender_2021 <- data.frame(
  SERIES_ID = c("One borrower: Male (33.5%)", "One borrower: Female (21.4%)", "Two borrowers (44.4%)",
  DECIMAL_VALUE = c(0.335, 0.214, 0.444, 0.006),
  label = paste0(round(c(0.335, 0.214, 0.444, 0.006) * 100, 1), "%")
)

# 1998 Pie Chart
p1998 <- ggplot(mortgages_borrowers_gender_1998, aes(x = "", y = DECIMAL_VALUE, fill = SERIES_ID)) +
  geom_bar(stat = "identity", width = 0.5, color = "black") +
  geom_text(aes(label = ifelse(rank(-DECIMAL_VALUE) <= 3, label, "")),
```

```r
            position = position_fill(vjust = 0.5),
            size = 3,
            fontface = "bold") +
  coord_polar(theta = "y") +
  labs(x = NULL,
       y = NULL,
       fill = "Number of Borrowers & Gender") +
  scale_fill_brewer(palette = "YlOrBr") +
  theme_void() +
  ggtitle("1998") +
  theme(legend.position = "left",
        plot.title.position = "plot",
        plot.title = element_text(hjust = 1)) +
  guides(fill = guide_legend(reverse = TRUE))

# 2021 Pie Chart
p2021 <- ggplot(mortgages_borrowers_gender_2021, aes(x = "", y = DECIMAL_VALUE, fill = SERIES_ID)) +
  geom_bar(stat = "identity", width = 0.5, color = "black") +
  geom_text(aes(label = ifelse(rank(-DECIMAL_VALUE) <= 3, label, "")),
            position = position_fill(vjust = 0.6),
            size = 3,
            fontface = "bold") +
  coord_polar(theta = "y") +
  labs(x = NULL,
       y = NULL,
       fill = "Number of Borrowers & Gender") +
  scale_fill_brewer(palette = "YlOrBr") +
  theme_void() +
  ggtitle("2021") +
  theme(legend.position = "left",
        plot.title.position = "plot",
        plot.title = element_text(hjust = 1)) +
  guides(fill = guide_legend(reverse = TRUE))

# Display both pie charts together
# grid.arrange(p1998, p2021, ncol = 1)
```